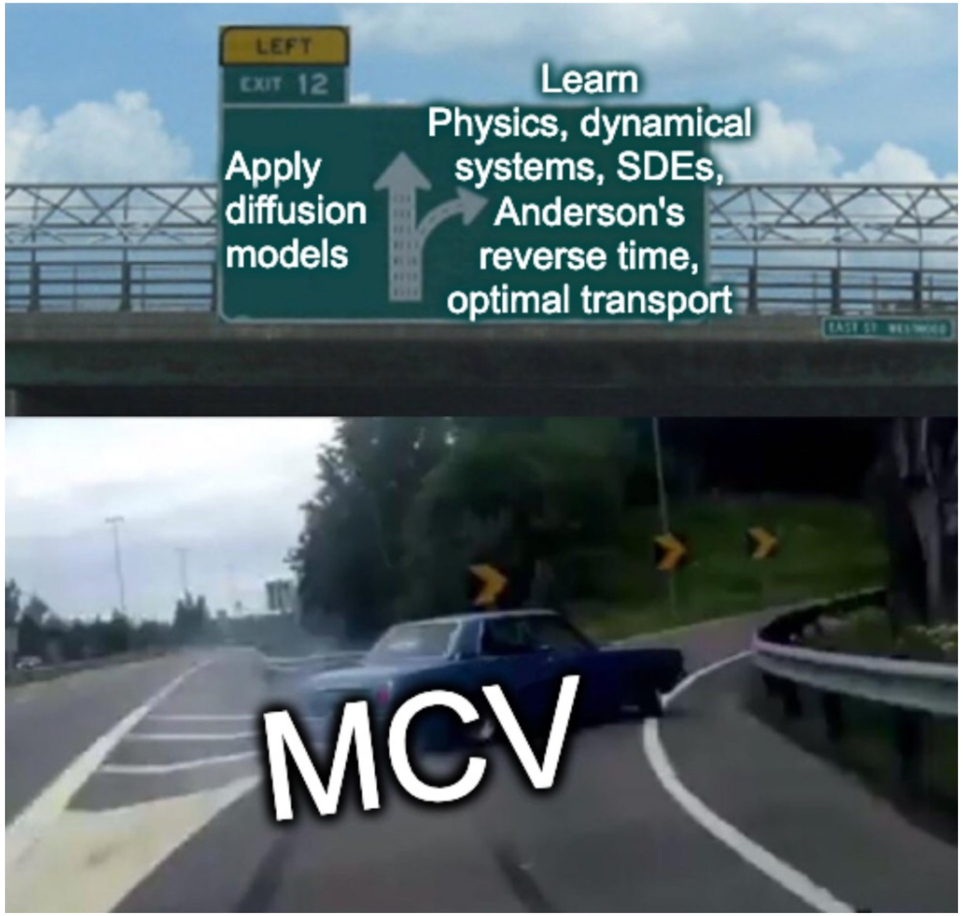
# Lecture 10: Generative Models (2)



# Today:

**00:00)** Dynamical systems crash course

**00:22)** Statistical mechanics crash course

**00:45)** ---- Break ----

**01:00)** Reverse diffusion

**01:20)** Score function and ELBO

**01:45)** ---- Break ----

**02:00)** Flow-matching

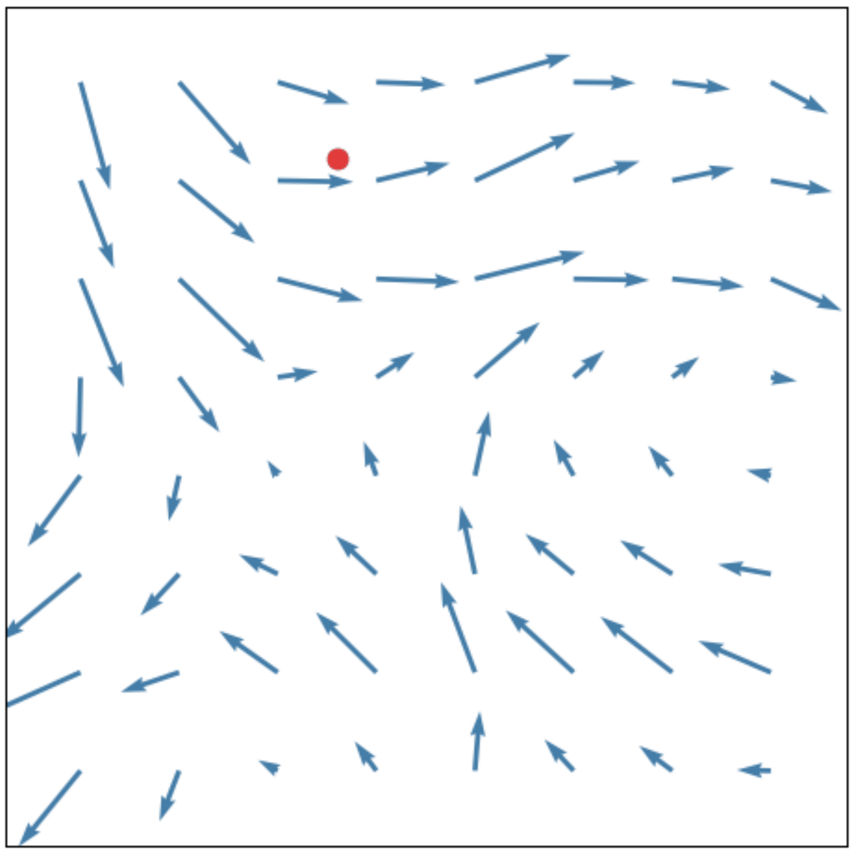
**02:15)** Variants: Consistency models, MeanFlow

**02:30)** Idempotent Generative Networks

**02:45)** ---- Break ----

**03:00)** Tutorial: Detection and Segmentation

# Dynamical systems crash course




- FNF: Fancy Name For

- Vector field: FNF $\boxed{f(\mathbb{R}^N) \to \mathbb{R}^N}$ (for our cases).

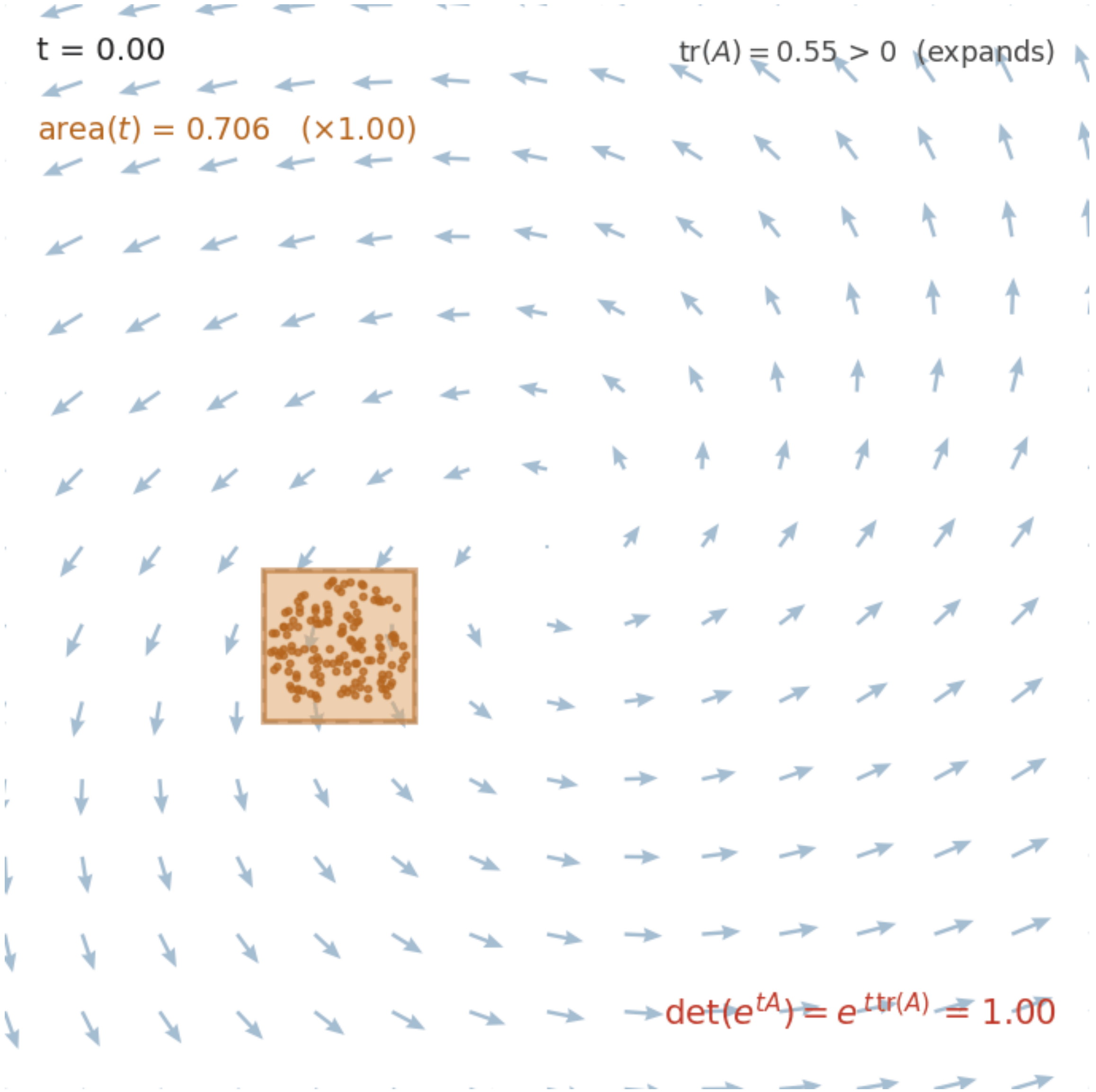
- Flow: FNF travelling on a vector field
  $\boxed{x_{t+\Delta t} = x_t + \Delta t \cdot f(x_t), \qquad \Delta t \to 0}$.
  **Numeric solution for the ODE: Euler step.**

- Velocity: FNF derivative of the flow w.r.t time
  $\boxed{v_t = \lim_{\Delta t \to 0} \frac{x_{t+\Delta t} - x_t}{\Delta t} = \frac{dx_t}{dt} := \dot{x}_t}$.

- So: $\boxed{f(x_t) = \frac{dx_t}{dt}}$. The flow is described by an ODE!

- Q: what's the solution if $f(\cdot)$ is linear?

# What about volume?




- Divergence $\boxed{\nabla f = \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} \ldots + \frac{\partial f_N}{\partial x_N}}$ can be understood in 2 ways:
  1. Flux out through tiny closed surface.
  2. The relative rate of change of volume.

- For a linear vec field $\boxed{f(x) = Ax}$, Div is the matrix trace:

$$\boxed{\nabla \cdot f = \sum_{n=1}^{N} \frac{\partial f_n}{\partial x_n} = \sum_{n=1}^{N} A_{nn} = tr(A)}$$

(For non-lin, it's trace of the Jacobian).

- Determinant is the volume multiplier (for linear field):

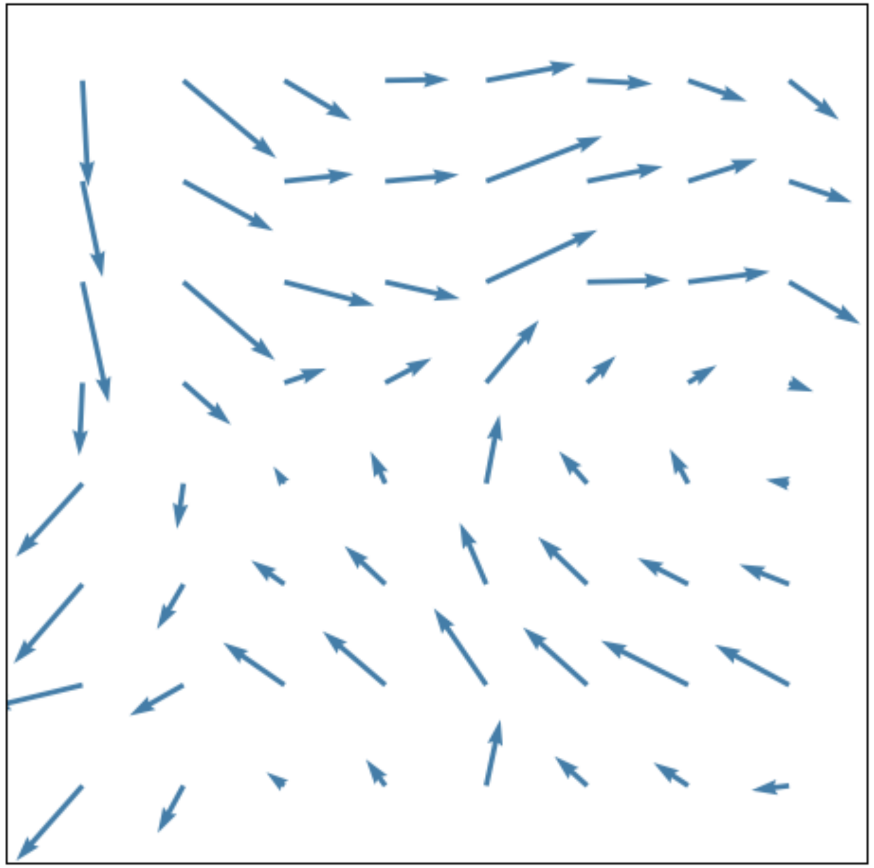
$$\boxed{\det(A) = {}^{AV_0}/_{V_0} . \Rightarrow \quad AV_0 = V_0 \det(A)}$$

- Recall linear field flow $e^{tA}$, so $\boxed{V_t = V_0 \cdot \det(e^{tA})}$

- But also $\nabla f = tr(A)$ is the infinitesimal progression of volume. so $V_0 e^{t \cdot tr(A)}$ is the scalar volume flow. And:

$$\boxed{\boldsymbol{\det\left(e^{tA}\right) = e^{t \cdot tr(A)}}}$$

# Statistical mechanics crash course




- We get random flow

$$\boxed{x_{t+\Delta t} = x_t + \Delta t \cdot f(x_t) + \sqrt{\Delta t \cdot \sigma_t^2}\; z_t, \quad z_t \sim \mathcal{N}(0,1) \quad , \quad \Delta t \to 0}$$

$\boldsymbol{Euler - Maruyama\ equation}$.

Each particle is doing a tiny random walk.

- Differential form: **Langevin equation**.

$$\boxed{dx = f(x)dt + \sigma_t dW}$$

$(W_{t+\tau} - W_t) \sim \mathcal{N}(0, \tau)$ Wiener process / Brownian motion.

- The particle-flow is random, the distribution-flow deterministic.

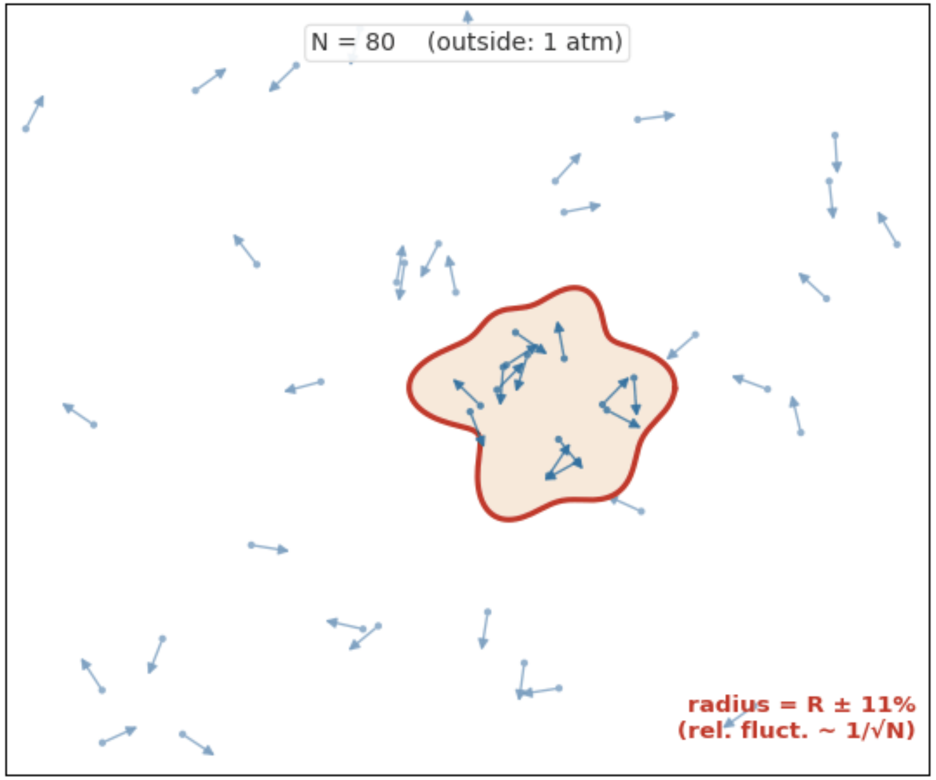
$$\boxed{\dot{p} = -\nabla \cdot (fp) + \frac{\sigma_t^2}{2} \nabla^2 p}$$ **Fokker-Planck equation.**

$$\nabla \cdot (\nabla p) = \nabla \cdot (p \underbrace{\boldsymbol{\nabla \log p}}_{\textbf{Score function}})$$

$$\boxed{\dot{p} = -\nabla \cdot p(f - \frac{\sigma_t^2}{2} S)}$$

# Is life just random?

Equilibrium: inside = outside (same density, speed)



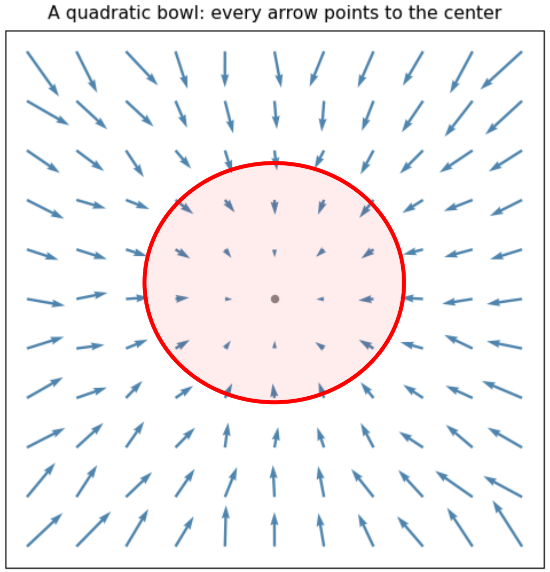

- Recall:
  The particle-flow is random, the distribution-flow deterministic.
  $$\dot{p} = -\nabla \cdot p(f - \frac{\sigma^2}{2}S)$$
  Inflation transient: $|f| < \frac{\sigma^2}{2}|S|$

- At equilibrium $pressure\_in = \ pressure\_out$ , $\dot{p} = 0$
  $$f = \frac{\sigma^2}{2}S$$
  What happens if outside is vacuum?

- But this is distribution flow!
  Boundary flux at each angle is ~Gaussian RV.
  (compound Poisson $\lambda \gg 1$), $r$ inherits it:
  $$\boxed{{}^{\sigma_r}/_{\mu_r} \sim {}^{1}/_{\sqrt{N}}}.$$

Nothing is deterministic, some things have low variance.

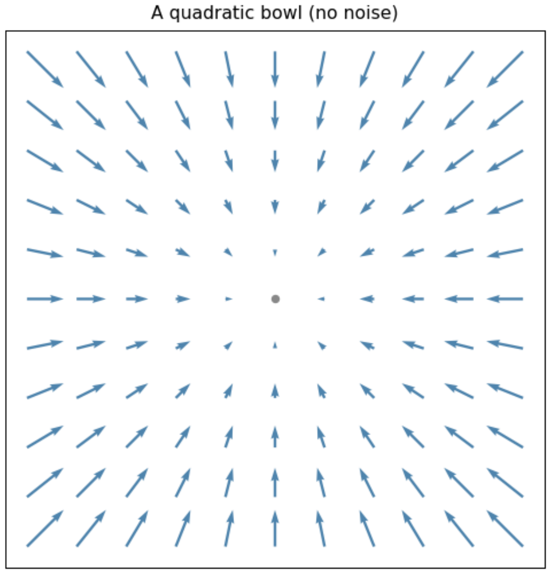
# DDPM noise schedule




$$dx_t = -\tfrac{1}{2}\beta_t x_t dt + \sqrt{\beta_t} dW$$

DDPM noise schedule,
Diffusion with drift,
Langevin dynamics.

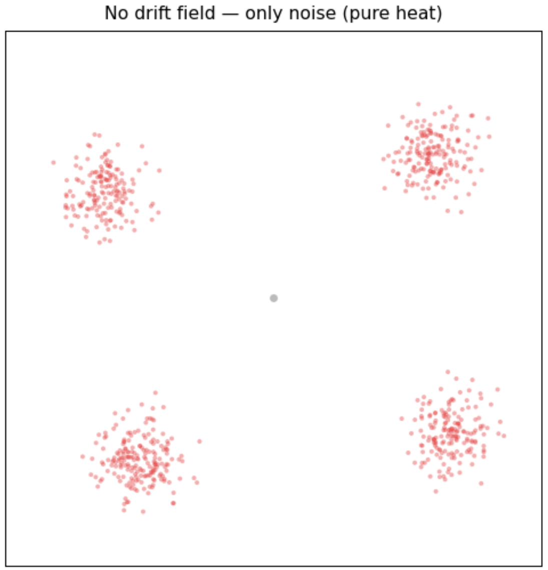
=




$$dx = -\tfrac{1}{2}\beta_t x_t dt$$

Drift,
Deterministic motion

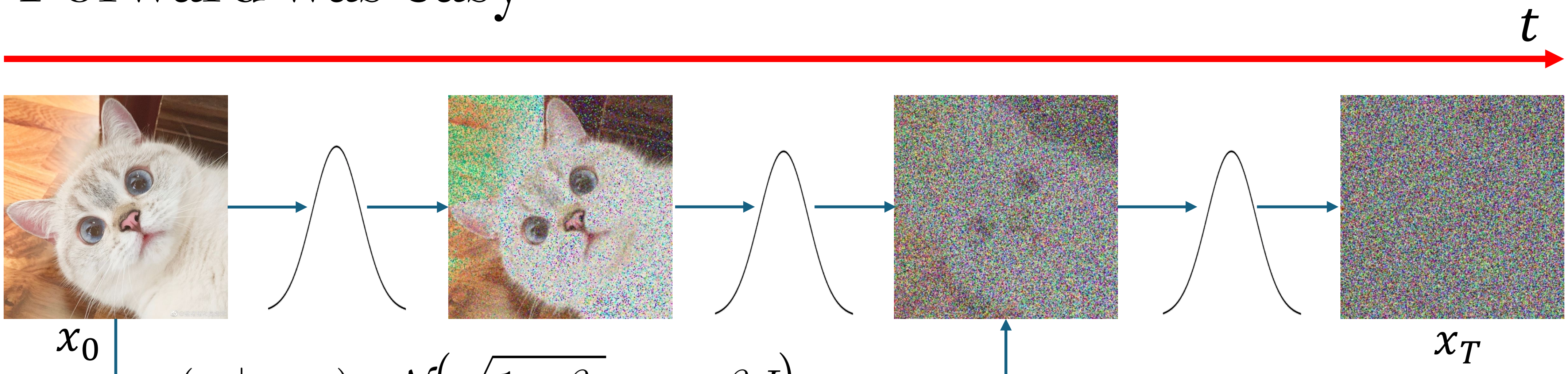
+




$$dx = \sqrt{\beta_t} dW$$

Heat / Diffusion.

# Forward was easy




$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\right)$$

$$q(x_t \mid x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I\right), \quad \bar{\alpha}_t = \prod_{s \le t} (1-\beta_s)$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$$


Bayes: $$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$ $\Longrightarrow$ Gaussian!

# Predict what?!

| | |
|---|---|
| **Noise (ε)** | |
| **Clean data ($x_0$)** | |
| **Velocity / flow field ($\frac{dx_t}{dt}$).** | E.g., FM: $x_1 - x_0$ |

# Reverse path

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I)$$

(Complete to square)

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t, \qquad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t$$

(Use a denoiser to predict)

$$\hat{x}_0 = \mathbb{E}[x_0 \mid x_t] \approx \mathrm{net}_\theta(x_t, t)$$

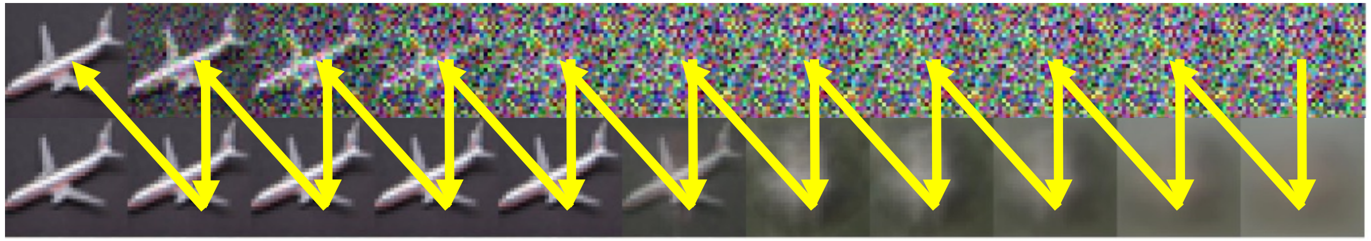
$$x_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0}_{\text{Mean drift}} + \underbrace{\left(\sqrt{1-\bar{\alpha}_{t-1}-\tilde{\beta}_t}\,\hat{\epsilon} + \sqrt{\tilde{\beta}_t}\,z\right)}_{\text{Noise: some old, some new}}$$

**DDPM sampling**

```
draw x_T ~ N(0, I)
for t = T ... 1:
    x0 = net(x_t, t)
    x_{t-1} = mu(x0) + noise
return x_0
```

$x_t$

$E[x_0|x_t]$



Inference

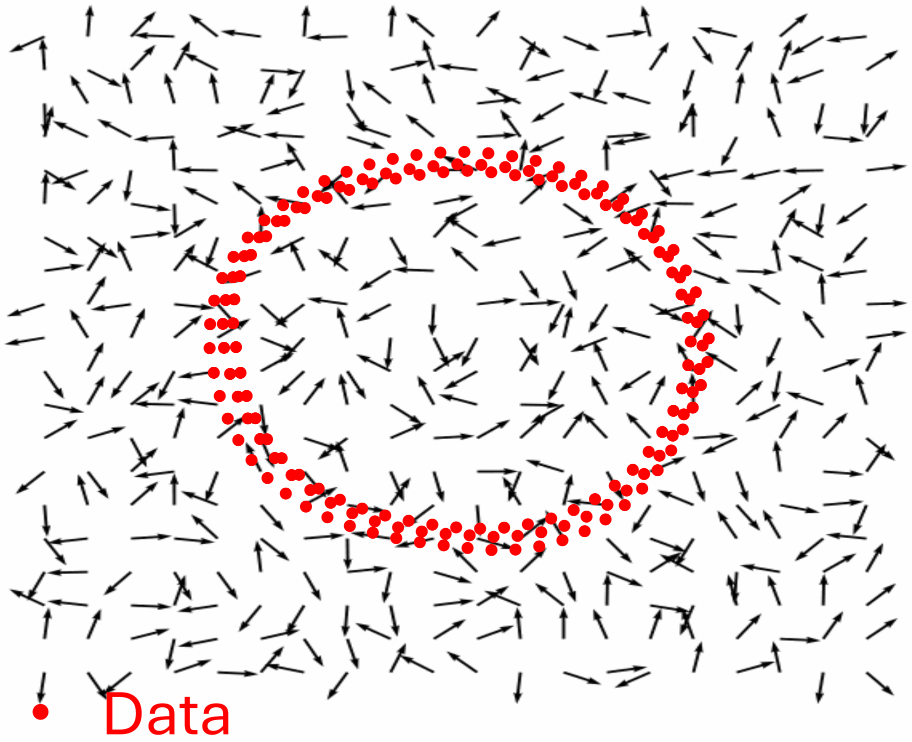
# It was the score all along

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\, \hat{x}_0 + \left( \sqrt{1 - \bar{\alpha}_{t-1} - \tilde{\beta}_t}\, \hat{\epsilon} + \sqrt{\tilde{\beta}_t}\, z \right)$$

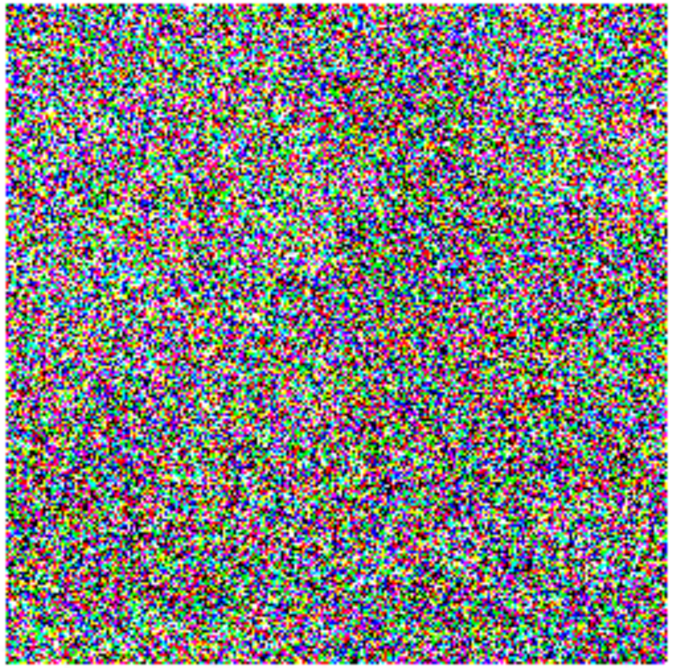

Tweedie's formula: For a Gaussian $y$, $\mathbb{E}[\mu \mid y] = y + \sigma^2 \nabla_y \log p(y)$

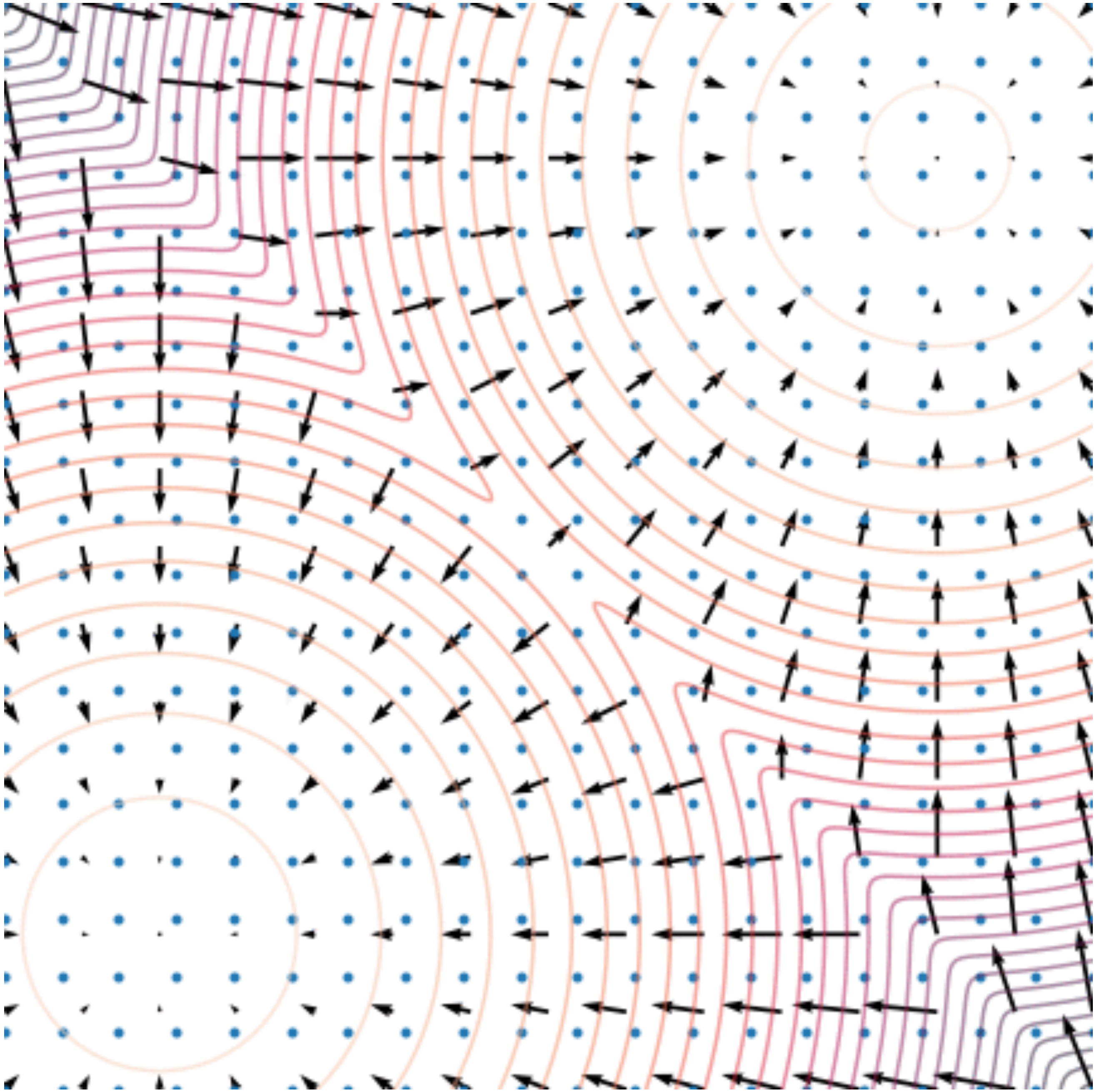
$$\hat{x}_0 = \mathbb{E}[x_0 \mid x_t] = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t + (1 - \bar{\alpha}_t)\, \underbrace{\nabla \log p_t}_{\textbf{SCORE}}\right)$$

$$x_{t-1} = \underbrace{x_t}_{\boxed{\text{Cur pos}}} + \underbrace{\tfrac{1}{2}\beta_t x_t}_{\boxed{\text{Reverse drift}}} + \underbrace{\beta_t S}_{\boxed{\text{Reverse diffusion}}} + \underbrace{\sqrt{\beta_t} z}_{\boxed{\text{Randomness}}}$$



# Sampling

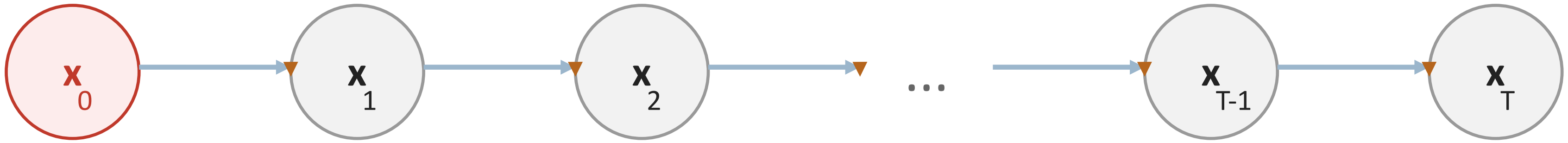




# FYI: DDPM is a deep VAE

**VAE**


**DDPM**




$$\log p_\theta(x_0) = \log \mathbb{E}_q\left[\frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right] \geq \mathbb{E}_q\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right] = \text{ELBO}$$

# Diffusion without diffusion? Going deterministic

DDPM

$$x_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0}_{\text{Mean drift}} + \underbrace{\left(\sqrt{1-\bar{\alpha}_{t-1}-\tilde{\beta}_t}\,\hat{\epsilon} + \sqrt{\tilde{\beta}_t}\,z\right)}_{\text{Noise: some old, some new}}$$

DDIM

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \underbrace{\sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}}_{\text{Noise: only old (predicted)}}$$

$$\tilde{x}_{t-1} = \tilde{x}_t + \hat{\epsilon}_\theta \cdot (\eta_{t-1} - \eta_t), \qquad \eta_t = \sqrt{\bar{\alpha}_t}/\sqrt{1-\bar{\alpha}_t}$$

Euler step!

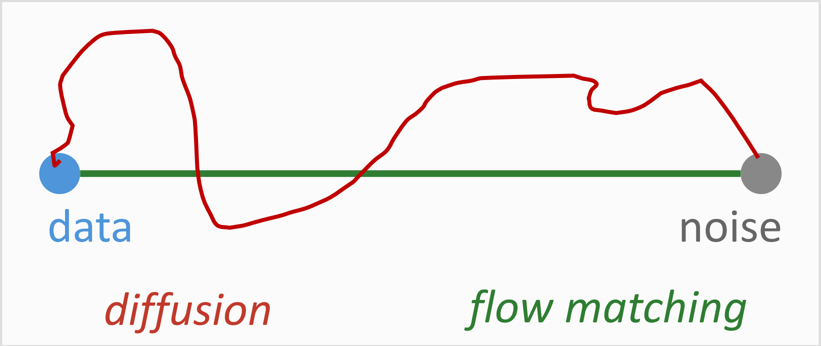
# Flow Matching

$x_1$: target clean image
$x_0$: instance of pure noise

Straight path: $$x_t = (1-t)x_0 + tx_1$$

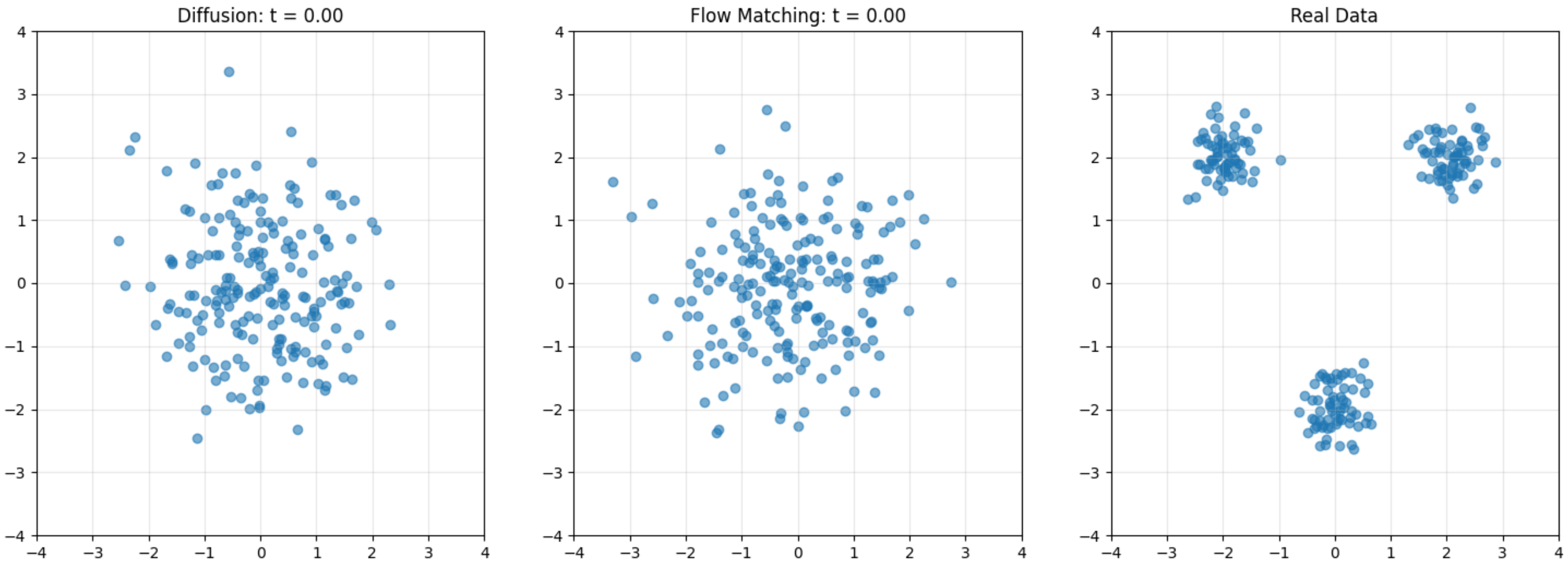
Note: $\dot{x}_t = x_1 - x_0$

**The vec field path from source to target is constant**

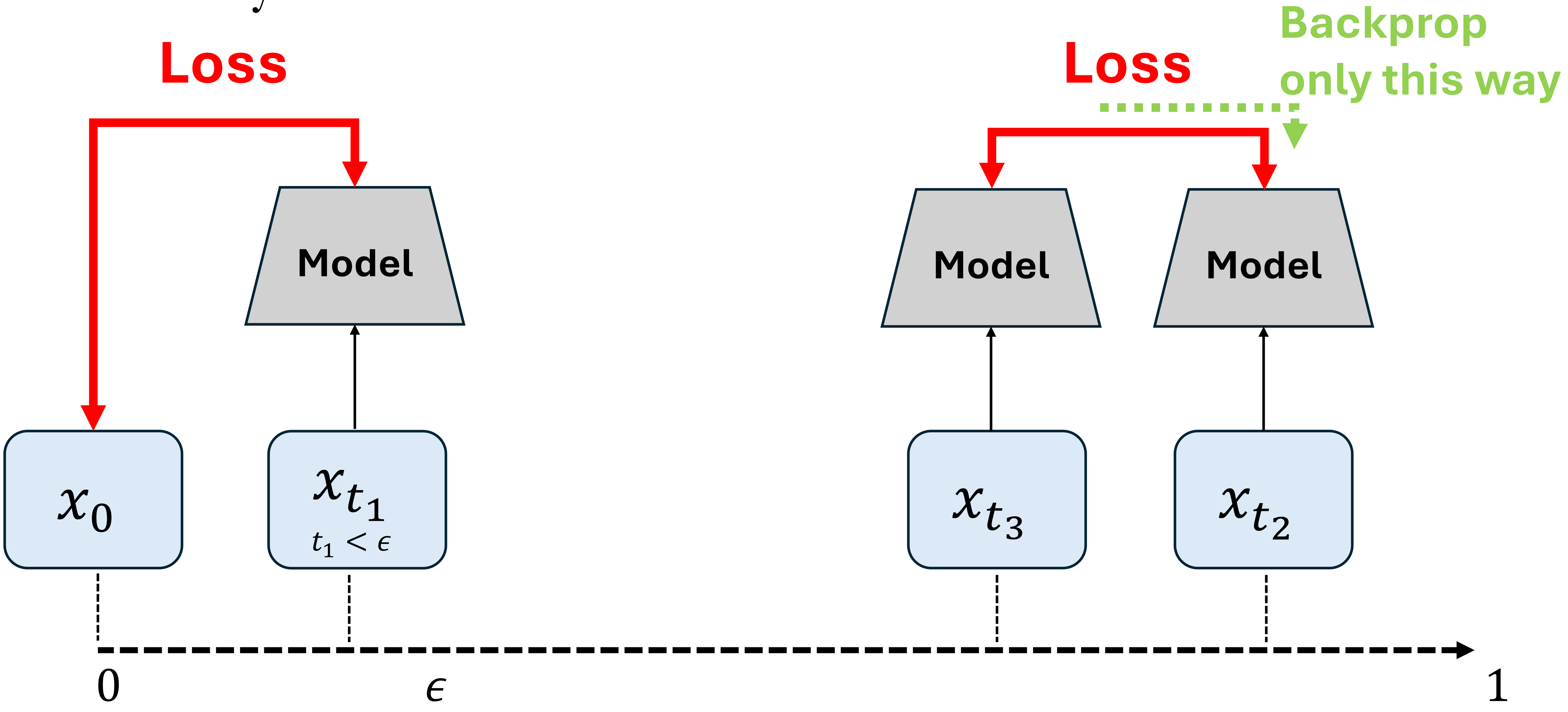


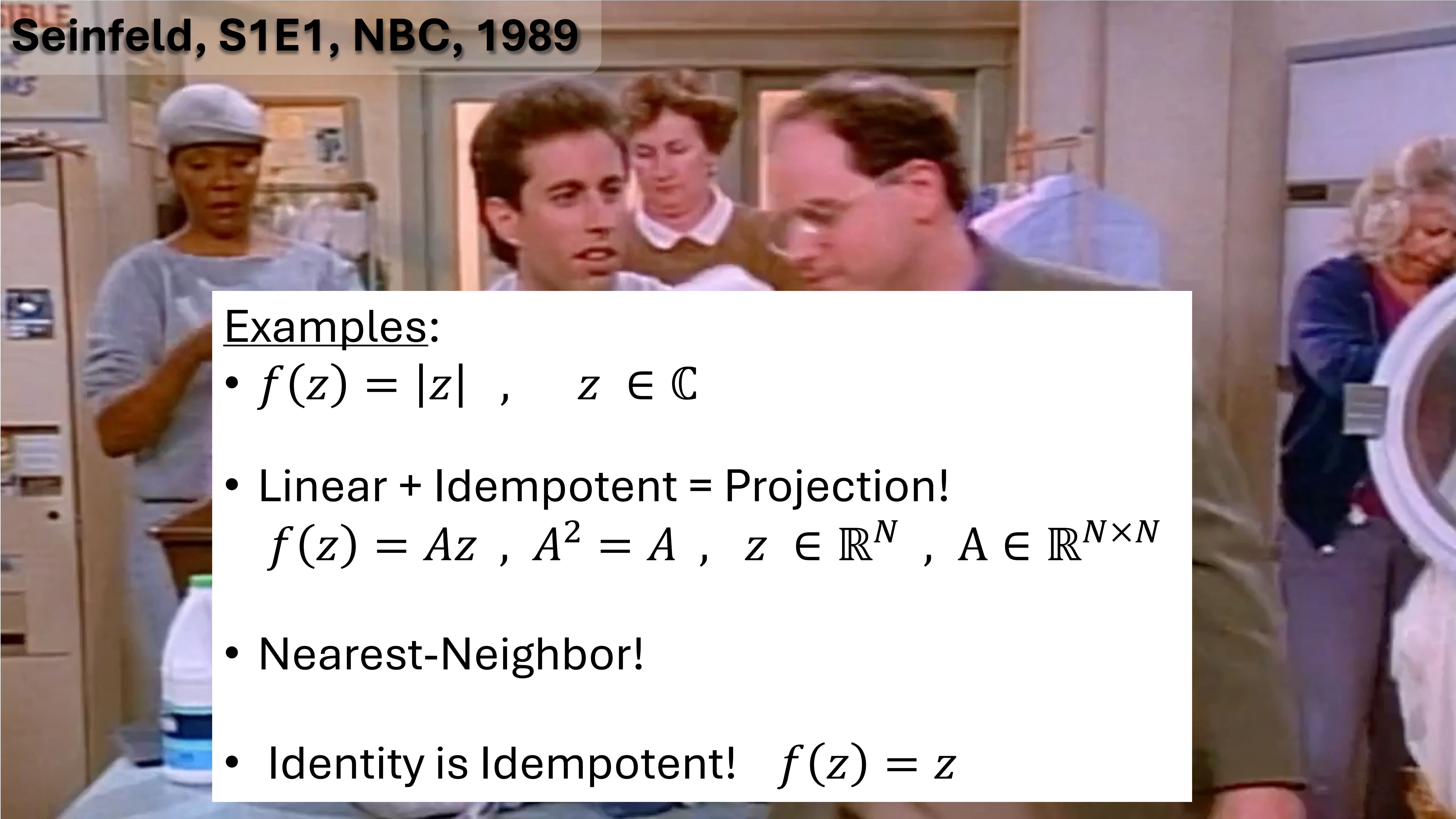

Predict velocity $\hat{V}_\theta(x_t, t) \approx \mathbb{E}[x_1 - x_0 | x_t]$

$$\mathcal{L}_{FM} = ||\hat{V}_\theta(x_t, t) - (x_1 - x_0)||$$



# Consistency Models





Seinfeld, S1E1, NBC, 1989

<u>Examples</u>:

- $f(z) = |z| \;\;,\;\;\;\; z \in \mathbb{C}$

- Linear + Idempotent = Projection!
  $f(z) = Az \;,\; A^2 = A \;,\;\; z \in \mathbb{R}^N \;,\; \mathrm{A} \in \mathbb{R}^{N \times N}$

- Nearest-Neighbor!

- Identity is Idempotent! $\;\; f(z) = z$

$$f(x) = x$$

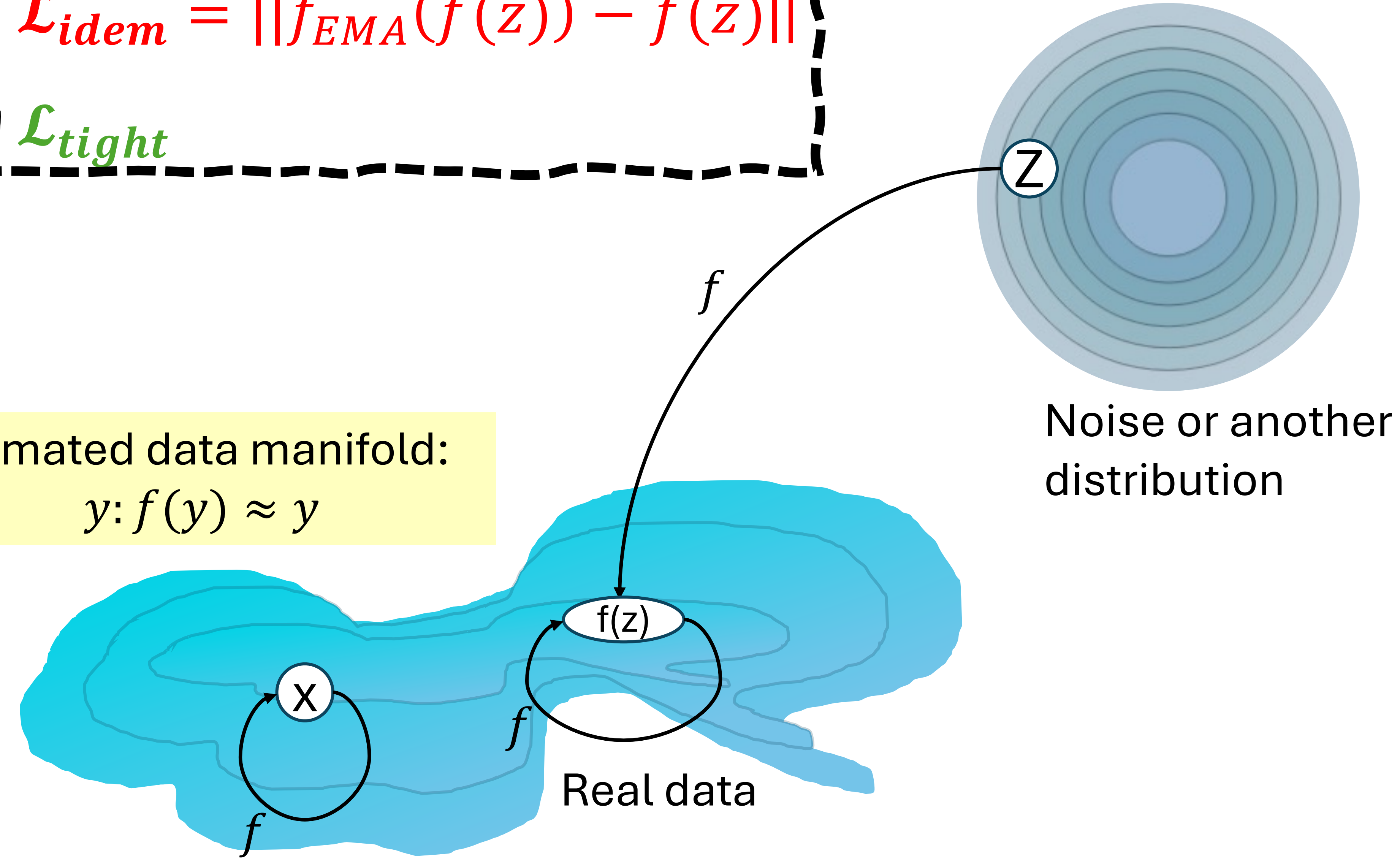
$$f(f(z)) = f(z)$$

$$\mathcal{L}_{rec} = ||f(x) - x||$$

$$\mathcal{L}_{idem} = ||f_{EMA}(f(z)) - f(z)||$$

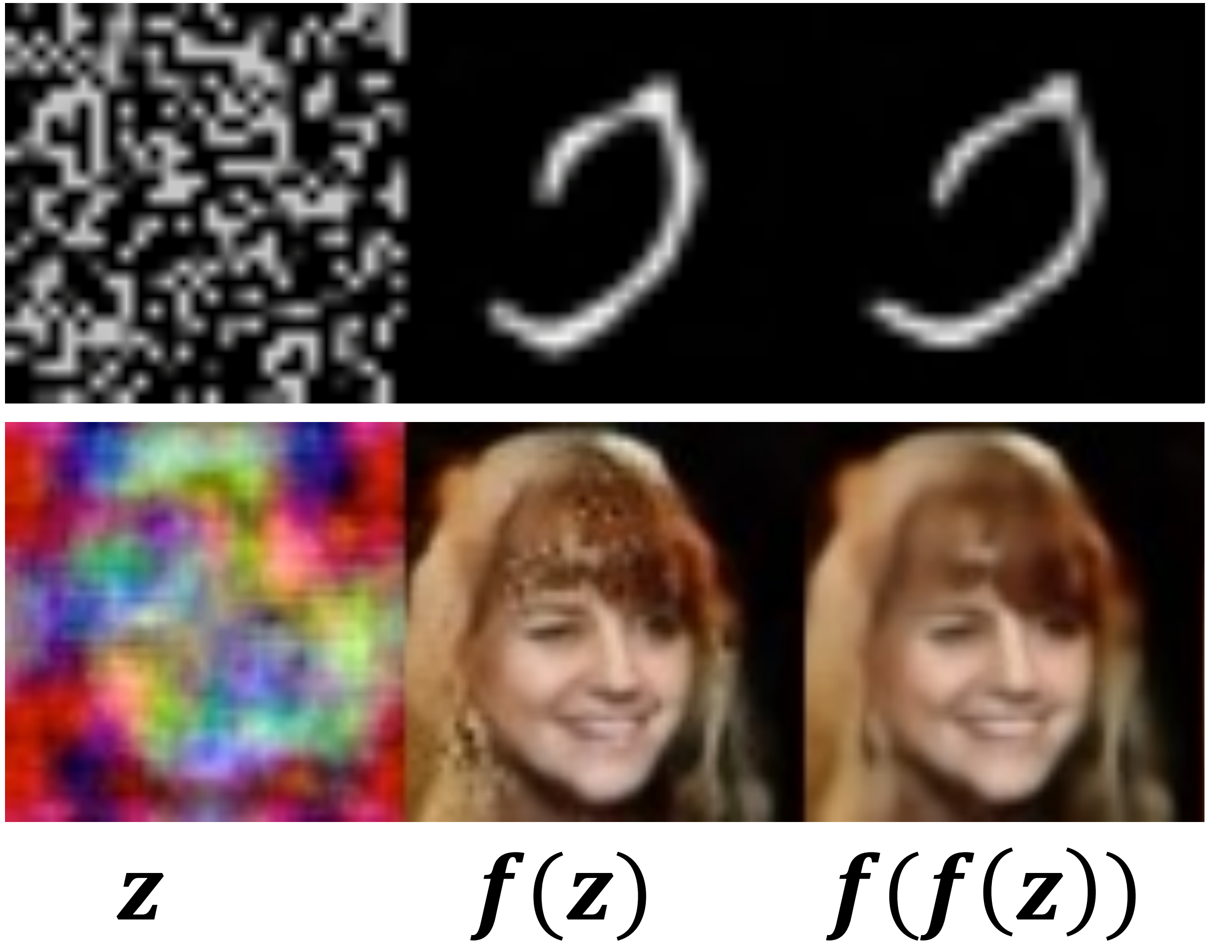
$$\mathcal{L}_{tight}$$

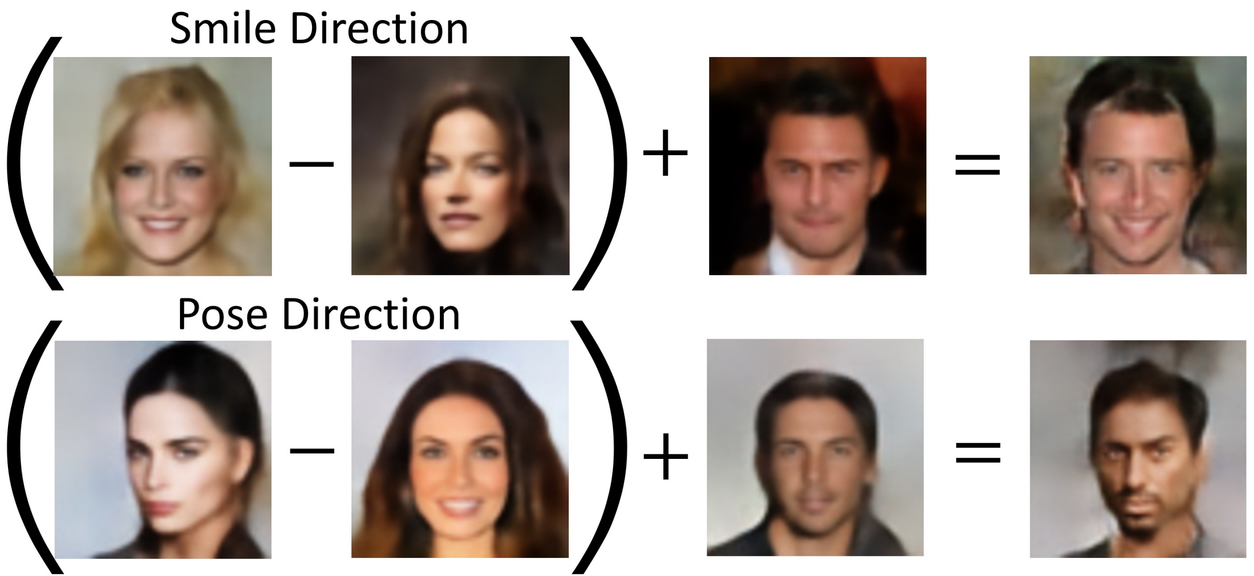
Estimated data manifold:
$y: f(y) \approx y$



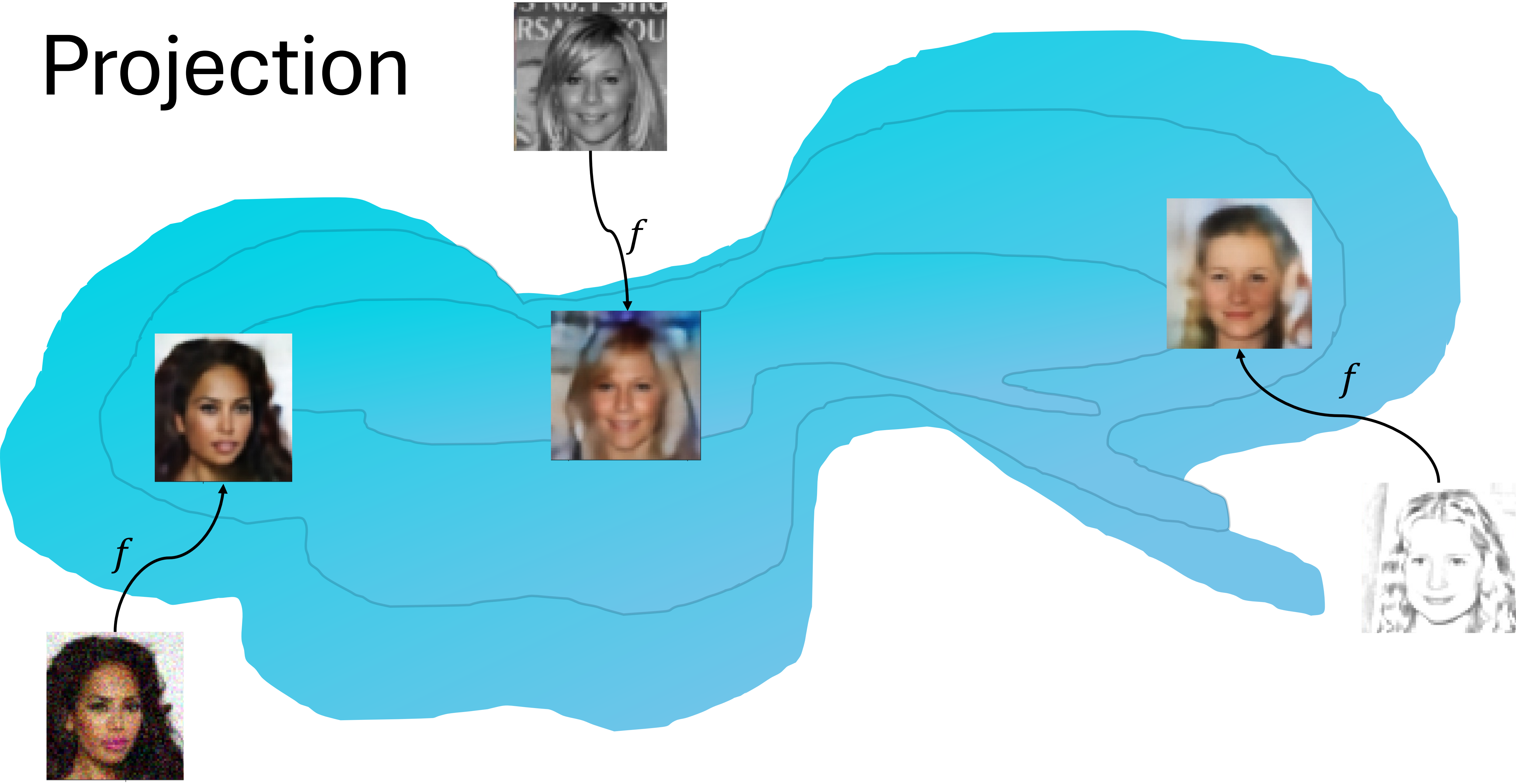

Shocher, Dravid, Gandelsman, Mosseri, Rubinstein, Efros, Idempotent Generative Network; ICLR’24

# Consistent Latent Space



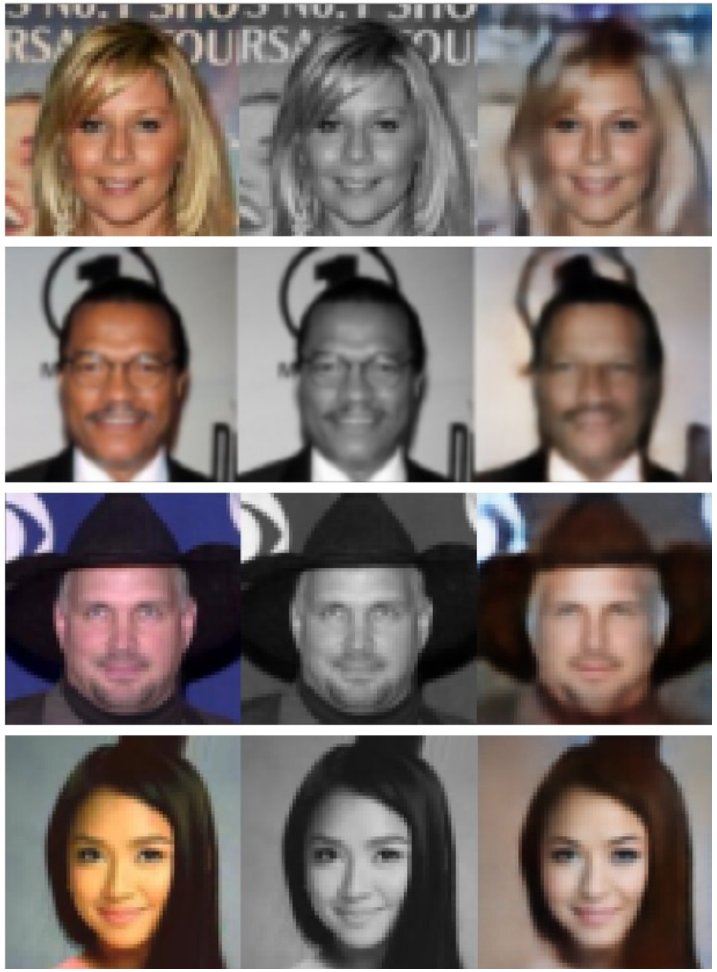



# Projection

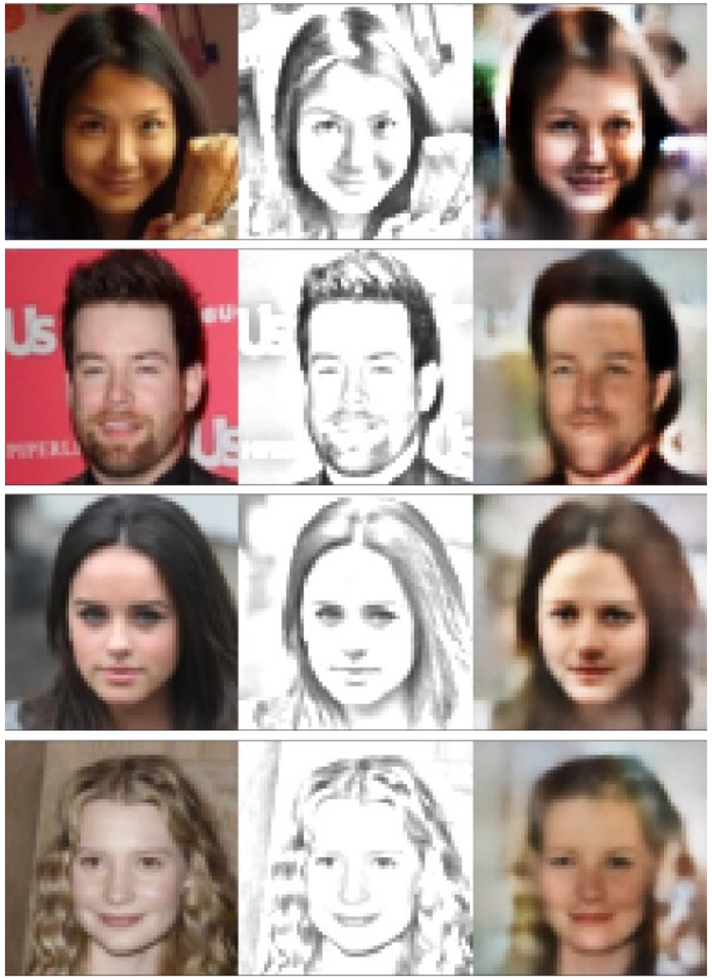



Shocher, Dravid, Gandelsman, Mosseri, Rubinstein, Efros, Idempotent Generative Network; ICLR'24

# Project to manifold



$x$ $\tilde{x}$ $f(\tilde{x})$



$x$ $\tilde{x}$ $f(\tilde{x})$

$x$ $\tilde{x}$ $f(\tilde{x})$

Shocher, Dravid, Gandelsman, Mosseri, Rubinstein, Efros, Idempotent Generative Network; ICLR’24

# Thank you!

## Next week:

Self-supervision & interpretability